

## Review

## The evolutionary history and distribution of cactus germplasm resources, as well as potential domestication under a changing climate

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**Abstract** The angiosperm family Cactaceae, a member of the Caryophyllales, is a large and diverse group of stem succulents comprising 1,438–1,870 species within approximately 130 genera predominantly distributed from northern Canada to Patagonia. Four centers of diversity from Central and North America (Chihuahua, Puebla-Oaxaca, Sonora-Sinaloa, and Jalisco) and three centers of diversity from South America (Southern Central Andes, Caatinga, and Mara Atlantica) have played a pivotal role in disbursing cacti around the globe. Mexico is considered the richest area in cacti species with close to 563 species grouped into 50 genera. Approximately 118 species have been domesticated by Mesoamerican people as food crops and for ornamental purposes. Cacti inhabit a wide range of ecosystems and climate regions, ranging from tropical to subtropical and from arid to semiarid regions. Species belonging to the genus *Opuntia* are the major food crop producers in the family. Cacti have derived characteristics from familial synapomorphies within the Caryophyllales. Reproduction occurs through pollination facilitated by birds, bats, bees, and other insects. Climate variability, whether natural or human-induced threatens global crop production including high temperatures, salinity, drought, flood, changes in soil pH, and urbanization. Cacti have several adaptations that are important for coping with abiotic stresses, such as crassulacean acid metabolism (CAM photosynthesis), as well as modifications to root and stem physiological pathways. This review aims to provide a comprehensive view of the fruit crops in Cactaceae, including their evolution, worldwide distribution, and the environmental factors impacting cultivation.

**Key words:** biotic and abiotic stresses, Cactaceae, edible fruits, evolution, germplasm resources, origin and distribution.

## 1 Cultivation: Definition and Emerging Challenges to Food Cultivation

A growing global population necessitates a vast amount of food (Shaw, 1945; Chen & Kates, 1994). Over the last half-century, one in every seven individuals has insufficient food for survival, and millions of people face starvation or take in insufficient quantities of proteins and vitamins in their diets (Godfray et al., 2010). Consistently, similar patterns of food habits and an increasing population have affected the food supply chain in multiple regions. The incorporation of

food items in products such as biofuels and cosmetics, as well as urbanization expectations and variability due to climate change, pose threats to global food security. (Parry & Lea, 2009; Tilman et al., 2011). Emerging challenges in the cultivation of food crops are influenced by several environmental factors. Climate change seriously threatens biodiversity, agriculture productivity, and food security (Morton, 2007; Reidsma et al., 2010; Vermeulen et al., 2012b; Wiebe et al., 2019; Chandio et al., 2021) either directly via causing changes in agro-ecological zones, especially productivity and patterns of crops, or indirectly by affecting growth and

social-economic development (Schmidhuber & Tubiello, 2007; Brown & Funk, 2008; Vermeulen et al., 2012a). Since the middle of the 19th century, the global climate has changed (Mann et al., 2009; Screen & Simmonds, 2010; Rohde et al., 2013) with the overall mean temperature increasing by 0.8°C since the 1850s and by the end of 21st century a further increase from 2°C to 7°C is expected (Allison et al., 2011; Wheeler & Von Braun, 2013). Global climate change is expected to reduce agricultural productivity by more than 10% by 2050 (Tai et al., 2014). However, the 2022 report shows a 10%–30% loss in wheat production in India (Sidhu, 2022). In Pakistan losses of 10% and 30% were recorded in wheat and mango production due to the heat wave in March and April 2022 (Bhatti, 2022), thereby drastically worsening global food production. Precipitation shifts have a larger influence on agricultural productivity than changes in temperature (Wassmann et al., 2009). Temperature impacts crop productivity on a long timescale and reduces the nutritional value of grain, and seed viability, and causes seed shrinkage. From 1832 to 2014, the atmospheric carbon dioxide (CO<sub>2</sub>) content rose from around 284 to 400 mg/L (Tans & Keeling, 2013; Wheeler & Von Braun, 2013). Global food security in the future will depend on the physiological processes of agricultural plants, which will be impacted by climate change including rising CO<sub>2</sub> levels (Tausz-Pösch et al., 2013). Climate change also impacts the availability of water, which impacts crop yield, putting extreme pressure on the hydrological cycle and demand for global water resources (Haddeland et al., 2014). Variations in the water cycle have a significant impact on agriculture, affecting established cropping patterns. Since the 1970s, an increase in global aridity has been recorded in Africa, South Asia, Southern Europe, and Eastern Australia due to low precipitation (Dai, 2011). Global aridity increased from 17% in the 1950s to 27% in the first decade of the 21st century (Thornton et al., 2014) resulting in decreased crop yield (Lobell et al., 2011).

Biotic and abiotic factors affect both growth and productivity by triggering physiological, biochemical, molecular, and morphological changes (Orcutt & Nilsen, 2000). Crop productivity can be reduced by 30%–70% due to drought stress (Kumaraswamy & Shetty, 2016). Heat stress impacts seed germination, photosynthesis, phenology, and reproductive biology (flowering, viability of pollen, and pollinator populations) (Antle & McGuckin, 1993). Chilling or cold stress experienced by plants in a range of 0–15°C leads to the loss of major crop productivity (Yadav, 2010). Flooding occurs frequently in many regions of the world due to high rainfall often causing damage to crop production and leading to soil erosion (Schmidhuber & Tubiello, 2007). The growth of various crop species in tropical and subtropical regions is affected by nonfreezing low temperatures and exhibits symptoms of lower germination, chlorosis, stunted seedlings, reduction in leaf expansion, wilting of leaves, and necrosis. When exposed to low temperatures, plants respond with changes in gene expression and protein synthesis (Sanghera et al., 2011). Low temperatures cause cellular dehydration and membrane injury by ice formation in the intercellular compartments that stop the cellular processes causing chlorosis, wilting, sterility, or even death (Levitt, 1980; Thomashow, 1999).

The world population and level of urbanization are increasing daily; the current global population is approximately

7.8 billion with an increase to 9.7 billion expected by 2050 (Schultz, 2018; Steduto et al., 2018). The current urbanization rate is unprecedented due to the growth of the global population and continuous rural-to-urban migration (Vermeulen et al., 2012a). Urban populations are projected to increase from 3.5 billion in 2010 to 6.2 billion by the end of 2050 (Angel et al., 2012). Currently, cultivated land covers approximately 25% of the world's terrestrial surface (Cassman et al., 2005), while urban land is expanding faster than urban populations in most parts of the world (Seto et al., 2010). The expansion of urban land modifies hydrology, biogeochemistry, habitat, land cover, and surface energy balance (Grimm et al., 2008). The rapid increase in population and urban expansion affects food cultivation, with more than 60% of irrigated croplands located close to urban areas, emphasizing the potential for land conflict between urban and agricultural purposes (Thebo et al., 2014). Over the last three decades, studies have shown high rates of urbanization and loss of cropland recorded globally with the most affected countries being China, Egypt, India, Turkey, and the United States of America (Chen, 2007; Ahmad et al., 2016). The urbanization process is often assessed using build-up areas, such as sports areas, golf courses, and gardens, reducing the available land for the cultivation of crops (Zasada et al., 2013; Plieninger et al., 2015). The majority of future urban population growth and urban area expansion is expected to occur in Africa and Asia (Seto et al., 2012). These rapid-growth areas have high rates of poverty and are potentially prone to a systemic disturbance in the food system (Puma et al., 2015; D'Amour et al., 2016).

## 2 Cacti as Iconic Succulents

Succulence is described as the ability to survive prolonged droughts in arid climates. Evolutionary modifications, known as the succulent syndrome (Edwards & Ogburn, 2012), include anatomical, morphological, cellular, and metabolic modifications (Gibson & Nobel, 1986). Cactaceae, within the Caryophyllales, is the most well known and largest succulent family possessing several adaptations to aridity (Oldfield, 1997). Some cacti have succulent roots, for example, *Cereus aethiops*, members of *Peniocereus*, and *Tephrocactus* (Mauseth & Stone-Palmquist, 2001; Stone-Palmquist & Mauseth, 2002); leaves, for example, *Opuntia*: or stems, for example, as in members of *Echinopsis*, *Gymnocalycium*, and *Notocactus* (Mauseth & Plemons, 1995; Mauseth & Plemons-Rodriguez, 1997, 1998). However, most cacti species show water-storing adaptations in the pith and cortex, with the stem usually thick and green, and possessing photosynthetic activities (Charles, 2014). Cacti are characterized by unique morphological characters including apical meristems of shoots organized in four different zones, shoots observed as areoles and sunken ovaries in the receptacles covered with bracts and areoles (Gibson & Nobel, 1986). Cactaceae includes 1,438–1,870 species belonging to approximately 130 genera (Hunt et al., 2006; Nyffeler & Eggli, 2010), many of which can store water in their tissues to avoid desiccation. Some species of cacti also serve as tropical fruit crops, with cacti being a low-cost alternative food source for humans and livestock, as well as being planted as ornamentals

(Estrada-Luna et al., 2008). The tribe Cacteae produces edible fruits commonly known as pitahaya (fruits of vine cacti; *Selenicereus*), pitaya (fruits of columnar cacti; *Cereus* and *Stenocereus*) or dragon fruit that are cultivated as food crops in Latin America (Nerd et al., 2002b). The most common and popular vine cactus is *Selenicereus undatus*, which is cultivated globally. The columnar cacti *Stenocereus queretaroensis* serves as the most popular food crop of Cactaceae (Nerd et al., 2002b). Another well known species in cultivation is *Opuntia ficus-indica* (L.) commonly known as the prickly pear (Fig. 1) (Nobel, 2002).

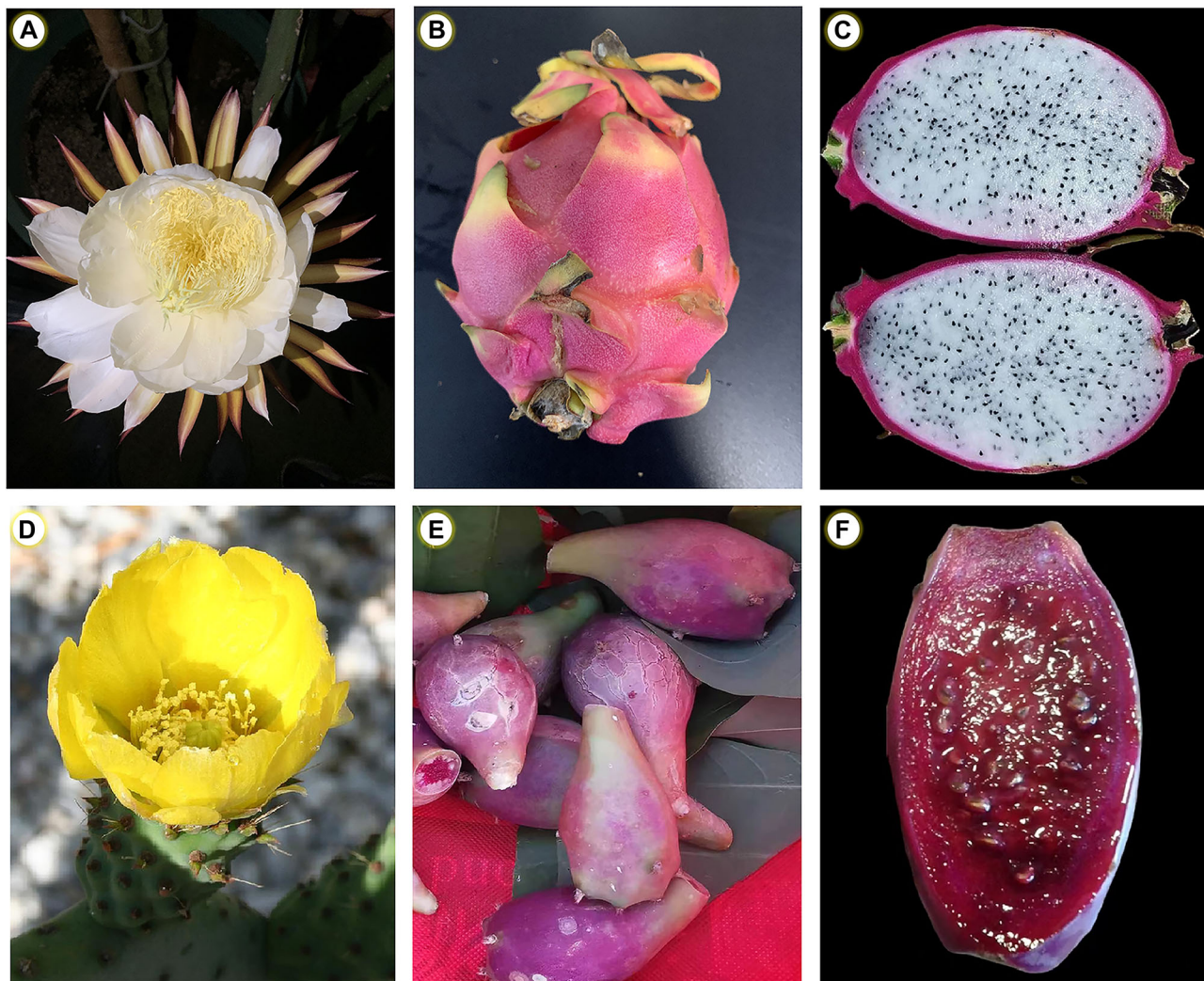
### 3 Evolution of Cactaceae

A poor fossil record for Cactaceae makes studying the evolutionary history of the family difficult (Arakaki et al., 2011). The ancestors of cacti were likely to have been leafy trees having a solid wood trunk, similar to the 17 extant species of *Pereskia* (Sajeva & Mauseth, 1991; Mauseth, 1995; Edwards &

Donoghue, 2006). Species of Cactaceae are found across the Western Hemisphere from southwestern Canada to Patagonia in Argentina and Chile. The epiphytic *Rhipsalis* has spread, presumably via birds, throughout tropical Africa, Madagascar, South India, and Sri Lanka (Anderson, 2001; Cota-Sánchez & Bomfim-Patricio, 2010; Majure et al., 2012). Cacti have commercial importance with various products being produced such as candies, ice creams, jams, jellies, marmalades, and syrups (Pimienta-Barrios & Nobel, 1994). To reconstruct the evolutionary history of Cactaceae there is a dire need to work on species identification, genetic classification, taxonomy, and speciation events through phylogenies.

### 4 Phylogeny of Cactaceae

Cactaceae belongs to the Caryophyllales, which also includes amaranths (Amaranthaceae), bougainvilleas (Nyctaginaceae), carnations (Caryophyllaceae), ice plants (Aizoaceae),



**Fig. 1.** Fruit and flower of *Selenicereus undatus* (Haw.) D.R.Hunt (syn. *Hylocereus undatus*) and *Opuntia ficus-indica* (L.) Mill. **A**, The open white flower. **B**, The red fruit with green scales. **C**, White flesh of *S. undatus* which contain many black seeds. **D**, The open deep yellow cup-shape flower. **E**, The red fruit. **F**, Deep red flesh of *O. ficus-indica*.

portulacas (Portulacaceae), pokeweeds (Phytolaccaceae), and saltbushes (Chenopodiaceae). The taxonomic classification of Cactaceae has been thoroughly reviewed by Cronquist & Thorne (1994), and the phylogenetic position of cacti as core members of Caryophyllales is universally accepted. Cacti have derived characteristics that differentiate them from other families within the order; hence within the Caryophyllales its phylogenetic placement is undisputed with other core members Aizoaceae, Amaranthaceae, Caryophyllaceae, Molluginaceae, Nyctaginaceae, Phytolaccaceae, and Portulacaceae (Cuénoud et al., 2002; Fig. S1). The presence of betalains, a type of nitrogenous pigments generated from tyrosine within Caryophyllales, is evidence of a chemical synapomorphy (Clement & Mabry, 1996). Proteinaceous plastid inclusion (designated as type P3cf.) is formed by members of Cactaceae and members of other closely related families during the ontogeny of sieve-tube members (Behnke, 1994). These unrelated and distinct synapomorphy features provide compelling evidence for the recognition of a monophyletic group.

Caryophyllales was classified using chloroplast DNA (cpDNA) restriction site mutations, which was validated by the deletion of the *rpl2* intron in the order's common ancestors (Downie & Palmer, 1994). Some unsolved concerns about the Caryophyllales and other orders remain such as with Laurales, Magnoliales, and Piperales, which have been shown by molecular studies to share close affinities to the Caryophyllales and should be classified within the order (The Angiosperm Phylogeny Group, 1998). This also includes the pitcher plants (Nepenthaceae) and insectivorous sundews (Droseraceae).

The phylogenetic relationship of Cactaceae within the Caryophyllales has been difficult to determine. Traditional and developmental comparative data have favored Phytolaccaceae (Buxbaum, 1953; Cronquist & Takhtadzhian, 1981) or Aizoaceae highlighting floral features (Turner, 1972). More recent studies have shown that Cactaceae and Portulacaceae (Gibson & Nobel, 1986; Gibson, 1994; Hershkovitz & Zimmer, 1997) have a recent common ancestor classified within the suborder Portulacineae Thorne (Cronquist & Thorne, 1994), which includes Basellaceae, Cactaceae, Didiereaceae, and Portulacaceae.

The monophyletic Portulacineae is strongly supported by gene sequences (Manhart & Rettig, 1994). The deletion of a 500 base pair (bp) fragment in the RuBisCO gene *rbcl* is recorded in certain Portulacaceae and Cactaceae, suggesting the two share a sister relationship (Rettig et al., 1992; Downie & Palmer, 1994). By studying the 1,100 bp open reading frame of the chloroplast genome locus ORF2280 (6.8 kb in size, largest gene in most plastid genomes), Downie et al. (1997) reported that *Pereskia* (Cactaceae) belongs to the portulacaceous cohort. The leaf-bearing ancient cacti were nested phylogenetically within Cactaceae and Portulacaceae, as the clade's sister taxon including *Talinum* by analyzing the internal transcribed spacer (ITS) ribosomal DNA sequence (Hershkovitz & Zimmer, 1997). An analysis was conducted using the chloroplast gene *ndhF* by Applequist & Wallace (2001) and concluded that Cactaceae was closely related to *Portulaca* and *Talinum* and nested within Portulacaceae *sensu lato*.

The monophyly and morphological distinctiveness of Cactaceae have been confirmed by molecular data of over

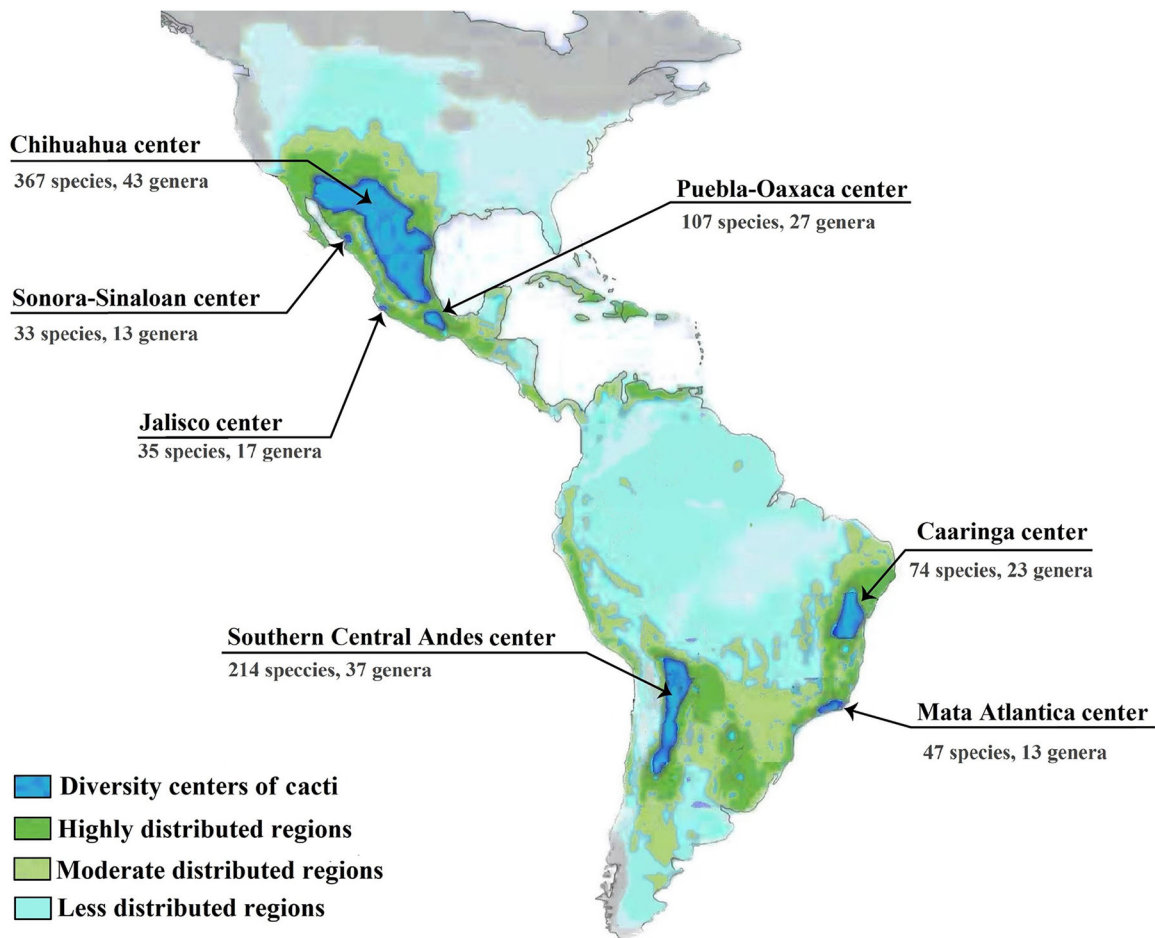
400 cacti species. A 6 kb inversion of the plastid genome involving *rbcl*, *atpE*, and *atpB* has been investigated in all cacti sampled, which is the best molecular synapomorphy to define Cactaceae (Wallace, 1995a, 1995b, 1997; Wallace & Cota, 1996). Within the Caryophyllales, Chenopodiaceae shows a similar independent inversion (Downie & Palmer, 1994). Based on this 6 kb chloroplast fragment, cacti systematists have concluded that Cactaceae is monophyletic, with a single ancestral population in which the inversion became fixed.

## 5 Historical (Nonanthropogenic) Biogeography of Cactaceae and Modern Distribution Centers

Biogeographic analyses of cacti are inhibited by only a few available fossils. Most researchers agree that Cactaceae originated in South America (Gibson & Nobel, 1986) and subsequently dispersed to Central America, North America, and the islands of the Western Hemisphere (Gibson & Nobel, 1986; Mauseth, 1990; Anderson, 1996; Stone-Palmquist & Mauseth, 2002). Barthlott et al. (2015) described seven centers of diversity for Cactaceae (Fig. 2), with various Cactaceae members having been distributed by anthropogenic activities over most of the temperate areas of the world (Benson, 1982). The largest numbers of cacti species are found in North America (Hernandez & Barcenas, 1995), with most found in the Chihuahuan Desert of Mexico which is considered the greatest center of diversity for Cactaceae. The western United States also has many cacti species, with the greatest concentration of taxa in California, Arizona, New Mexico, and Trans-Pesco Texas (Benson, 1982). The geographic distribution of cacti, in general, is limited by low temperatures, although some species are hardy enough to occupy northern latitudes that experience cold temperatures, including the northern tier of the United States and Canada. The most extreme range of cacti is reported in the Peace River District of Alberta, Canada. Columnar cacti, with large white flowers, are reported to be found in Argentina, Bolivia, Chile, Ecuador, and Peru (Yetman, 2007). Brazil is considered the home of several cacti species in *Pilosocereus*, *Cereus*, and *Facheiroa* (Yetman, 2007).

The vegetation of cacti is important in South America including some tropical forests and the Caribbean, producing diverse agricultural products. Dense forests of cacti are found in the islands previously making up the Netherlands Antilles (Aruba, Bonaire, Curaçao, Saba, and Sint Eustatius). From the Andes to the Brazilian sertão to the desert of Aruba and the lushness of Cuba, giant cacti produce lumber and fruits. Horticulturalists of the dry Andes cultivate *Trichocereus atacamensis* for food and lumber. Species of *Cereus* grow wild in the humid regions of South America (Yetman, 2007). More than 130 species of columnar cacti are native to the Western Hemisphere, including the West Indies. *Rhipsalis* is an epiphytic cactus, native to tropical Africa as well as to the Western Hemisphere, but was most likely introduced by birds or Europeans via the sticky seeds (Yetman, 2007).





**Fig. 2.** Map of cacti diversity in four Central American centers; Chihuahua center (367 species, 43 genera), Sonora-Sinaloa center (33 species, 13 genera), Puebla-Oaxaca center (107 species, 27 genera), and Jalisco center (35 species, 17 genera) and three Southern America centers; Southern Central Andes center (214 species, 37 genera), Caatinga center (74 species, 23 genera) and Mata Atlantica center (47 species, 13 genera). Modified from Barthlott et al. (2015).

## 5.1 Central and North American centers of diversity

### 5.1.1 Chihuahua center

Chihuahua, located in Mexico, covers the largest area and comprises the largest number of species including 367 species within 43 genera. This center is a large complex of deserts and xeric shrublands which includes the Mexican Meseta Central Matorral. The richest diversity of globular cacti species of the tribe Cactaceae is found here and is considered as the main center of Opuntioideae species, while other species are absent or poorly represented. Most cacti species found here are shrubby and arborescent, with no evidence of the presence of columnar cacti.

### 5.1.2 Puebla-Oaxaca center

This primarily includes the Tehuacan-Cuicatlan valley, which has 107 species in 27 genera. This center is geographically separated from the Chihuahua center by the Trans-Mexican volcanic mountain belt. The vegetation types are xeric shrublands, dry tropical forests, thorn shrubs, and desert vegetation. This center has the highest columnar cacti species in Mexico. The species richness in this center is smaller than in the Chihuahua center.

### 5.1.3 Sonora-Sinaloa center

The area of this center is smaller, having a large number of cacti species including 22 species in 13 genera at the southern transition of the Sonoran Desert to the Sinaloa dry forest characterized by deciduous thorn forests. The region is characterized by a mixture of tropical and temperate cacti species. *Ferocactus* and *Mammillaria* are globular species that are present and the columnar *Stenocereus* are also found here. Phyllocactae subtribe *Echinocereinae* and shrubby *Opuntia* are other columnar cacti that can be found in this region.

### 5.1.4 Jalisco center

This is the smallest center of Central and North America having a high cactus diversity with 35 species in 17 genera. The climate of Jalisco is more humid than other diversity centers in the northern region and forms tropical dry forest vegetation. The cacti species found here are globular *Mammillaria* species, columnar cacti species, and arborescent cacti species such as *Cephalocereus*, *Stenocereus*, and *Opuntia*. Epiphytic cacti species like *Epiphyllum* and *Selenicereus* are also found in this center.

## 5.2 The Three South American centers of diversity

### 5.2.1 Southern Central Andes center

The largest South American center has the highest number of cacti species, represented by 214 species in 37 genera characterized by high morphological and phylogenetic diversity. This center is found in the arid and semiarid valleys of central and southern Bolivia, as well as northern Argentina. The vegetation in the northern zone of this center consists of deciduous woods with scattered naturally forest-free slopes with rock outcrops. In the southern Andean shrublands, semideserts, and the prepuna ecoregion are present. South America is rich in globular cacti, as well as the center of diversity for columnar cacti. The western sector of Bolivia's Tarija province and the bordering districts of Jujuy and Salta in northern Argentina possess exceptional variety and species richness.

### 5.2.2 Caatinga center

This Brazilian center of cacti diversity has 74 species in 23 genera and is located in the southern portion of the Caatinga ecoregion and the Campos Rupestres, which are adjacent to mountainous regions of the Chapada Diamantina and Serra Espinhaco. The main vegetation types of the Chapada Diamantina are xeric shrubland and thorn woodland. The highest cacti diversity is present in azonal habitat islands with rock outcrops or sandy soils with poor nutrients. The species diversity of this center is much lower than the Chihuahuas center and the Southern Central Andes center. The Caatinga center is home to the tribe Cereceae. The diversity of columnar species is high whereas the diversity of globose cacti is moderate.

### 5.2.3 Mata Atlantica center

This center is located in southeast Brazil having 47 species in 13 genera and is ecologically distinct from all other centers, characterized by rainy and humid evergreen forests. The prominent taxa of this center are arborescent species, for example, *Pereskia frandifoliai*, and especially the high number of epiphytic cacti species of *Phipsalis* and closely related genera. Other species can also be found in this center in isolated azonal habitats such as rock outcrops or coastal dunes.

## 6 An Inventory of Cacti Cultivated for Food Including Their Geography and Distributions

### 6.1 Summary statistics

Cacti are important fruit crops and have great potential as a horticultural crop. Nerd & Mizrahi (2010) reviewed the reproductive biology of cacti, which includes the attractive and unique fruits produced by many species in various genera. Most edible fruits, ranging in size from medium to large, belong to members of the subfamily Cactoideae (Table 1).

### 6.2 Major food crops of Cactaceae

Varieties of pitaya (also known as dragon fruit) are popular, refreshing, and visually appealing fruits that are sold widely in markets, especially in tropical and subtropical areas

(Chen et al., 2021a; Zaman et al., 2023). The native range of pitaya is believed to be in Mexican, Central American, and South American tropical forests (Mizrahi et al., 2010). Globally, the species has naturalized populations found in riparian zones, deciduous forests, coastal forests, tropical semideciduous forests, and thorn forests (Arias, 1997; Pérez-García et al., 2001). Dragon fruit can also be found in highly disturbed areas such as along roadsides, within maritime scrub, and among rocky terrain (Taylor & Zappi, 2004; Flora of China Editorial Committee, 2014).

### 6.2.1 *Selenicereus undatus* (Haw.) D.R.Hunt

Currently 15 *Selenicereus* species are recognized and are native to Costa Rica, El Salvador, the Pacific coast of Guatemala, and southern Mexico (IUCN, 2016). *Selenicereus undatus* (synonym *Hylocereus undatus* (Haw.) Britton & Rose; Tropicos.org) is commonly cultivated and distributed widely across the tropics and some temperate regions including Africa, Australia, China, Israel, North America, Southeast Asia, tropical regions of South America, and the West Indies (Fig. 3) (Nerd et al., 2002b; Taylor & Zappi, 2004; Dana et al., 2005a; Broome et al., 2007; Guo et al., 2014; Seebens et al., 2017; Witt, 2017; Zheng et al., 2018).

*Selenicereus undatus* fruits have a striking appearance with bright skin covered with green color scales and white flesh containing many tiny black seeds (Fig. 1). The flesh of the fruit is juicy with fruits up to 6–12 cm long, 4–9 cm thick, and oblong to ovoid shape with an average weight of 350–400 g (Figs. 1B, 1C). The flowers are known as “Queen of the Night” or “Noble Woman” referring to their beauty and scent. The nocturnally blooming flowers are hermaphroditic range in size from 25 to 30 cm, and are 17 cm wide. The flowers are bell-shaped and contain white stamens, and a cream-color lobed stigma (Fig. 1A). Plants are xerophytic vine-like and fast-growing. Stems are triangular, and sometimes four-sided or five-sided, fleshy, green in color, and considerably branched. Each stem segment consists of three wavy flat ribs and corneous (hornlike) margins that may have one to three small spines or a lack of spines. Stems are creeping, scandent, clambering, or sprawling up to 10 m long. Aerial roots arise from the underside of the stem which are capable of absorbing water and minerals from the soil (Merten, 2003).

*Selenicereus undatus* is a lithophyte plant (can grow on bare rocks or stones) or hemi-epiphyte (can grow on other plants), shade tolerant and drought resistant supported by crassulacean acid metabolism (CAM) (Andrade et al., 2007). The species grows in tropical semideciduous forests, coastal forests, tropical deciduous forests, riparian vegetation, thorn scrub, and thorn forests (Arias, 1997; Pérez-García et al., 2001). *Selenicereus undatus* can also grow in disturbed habitats, such as along roadsides, rocky outcroppings, and marine scrub (Taylor & Zappi, 2004; Flora of China Editorial Committee, 2014). The species can be found in widely diverse habitats of Mexico ranging from two to 2750 m above sea level, with 340–3500 mm annual rainfall and annual mean temperatures ranging from 13°C to 29°C (Cálix, 2004).

### 6.2.2 *Opuntia ficus-indica* (L.) Mill

*Opuntia ficus-indica* has a large, segmented trunk, ranging in height from 5 to 7 m, a crown over 3 m in diameter, and a trunk up to 1 m in diameter. A green or blue-green stem is

**Table 1** Important cacti fruits, their origin and distribution

| Species                                                                    | Common names                                                              | Native countries                           | Distributed to                                                                                                                                                                                                          | Fruit characteristics                                                                                                                                                                                                                     |
|----------------------------------------------------------------------------|---------------------------------------------------------------------------|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Selenicereus costaricensis</i> (F.A.C.Weber) S.Arias & N.Korotkova      | Pitahaya, pitahaya roja, red-purple pitahaya                              | Mexico and Central America                 | Mexico, Central America, Nicaragua, Guatemala, Panama, Costa Rica, Cambodia, Thailand, China, Malaysia, Vietnam, Sri Lanka, Philippines, Indonesia and Bangladesh, Okinawa, Hawaii, Israel, North Australia, and Cyprus | The fruit is scarlet with diameter of 10–15 cm and with weight 250–600 g. The fruit is ovoid, covered by scale, and its flesh is red-purple which contains number of small black seeds                                                    |
| <i>Selenicereus monacanthus</i> (Lem.) D.R.Hunt                            | Pitahaya, pitahaya roja, red pitahaya<br>Red Eden (Israel)                | Mexico                                     | Vietnam, China, Israel, Thailand, Australia, the United States, and Malaysia                                                                                                                                            | The fruit is large oblong with diameter 10–12 cm and weight 130–350 g which is covered with scales, the fruit flesh is red-purple, containing small black seeds                                                                           |
| <i>Selenicereus purpusii</i> (Weing.) S.Arias & N.Korotkova                | Pitahaya, purple dragon fruit                                             | Mexico                                     | America and Mexico                                                                                                                                                                                                      | The fruit is oblong with diameter 10–15 cm and weight 150–400 g having large scales and the flesh is red containing many small black seeds                                                                                                |
| <i>Selenicereus undatus</i> (Haw.) D.R.Hunt                                | Pitahaya, pitahaya roja, red pitahaya, dragon fruit (Asia), Eden (Israel) | Southern Mexico and Central America        | Bahamas, Bermuda, Indonesia, Colombia, Israel, Philippines, Myanmar, Malaysia, Mexico, Nicaragua, northern Australia, Okinawa (Japan), Sri Lanka, southern Florida, China, Thailand, Vietnam, and the West Indies       | The fruit is oblong with the length of fruit ranges from 15 to 22 cm and weight 300–800 g. The fruit is rosy-red and the scales are long and large having red or green tips. The flesh is white which contain small size many black seeds |
| <i>Selenicereus trigonus</i> (Haw.) S.Arias & N.Korotkova                  | Strawberry pear                                                           | United States (Puerto Rico; Virgin Island) | United States (Puerto Rico; Virgin Island)                                                                                                                                                                              | Fruits are ovoid or oblong (7–9 cm in diameter and 120–250 g in weight). Fruit scales are red and flash is white having many small black seeds. (not very pronounced flavor)                                                              |
| <i>Cereus peruvianus</i> (L.) Miller (Cereus jamacaru, Cereus uruguayanus) | Pitaya, apple cactus, Koubo (Israel)                                      | South American countries                   | Israel, United States (California), State of Rio de Janeiro, Brazil, the Netherlands, Antilles                                                                                                                          | Fruit is oblong and medium-large in size having smooth peel, spineless, usually red in color and the pulp is white                                                                                                                        |
| <i>Stenocereus griseus</i> (Haworth) Buxbaum                               | Pitaya, pitayo de Mayo                                                    | Mexico                                     | Venezuela, Colombia, ABC islands                                                                                                                                                                                        | Fruit skin is green or red and pulp is reddish                                                                                                                                                                                            |
| <i>Stenocereus queretaroensis</i> (Weber) Buxbaum                          | Pitaya, pitaya de Querétaro                                               | Mexico, state of Querétaro                 | Mexico, America states                                                                                                                                                                                                  | Fruit is globose, having spines, peel is red or green and the pulp is white or purple                                                                                                                                                     |
| <i>Stenocereus stellatus</i> (Pfeiffer) Riccobob                           | Pitaya, pitaya de Augusto                                                 | Arid and semiarid regions of America       | Distributed from South Arizona to northern Colombia and Venezuela, Mexico                                                                                                                                               | Fruit is globose having spines on peel, green or red in color and the pulp is colored or white                                                                                                                                            |

Continued

Table 1 Continued

| Species                               | Common names                                                                   | Native countries                                                   | Distributed to                                                                                                   | Fruit characteristics                                                                                                       |
|---------------------------------------|--------------------------------------------------------------------------------|--------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <i>Opuntia ficus-indica</i> (L.) Mill | Prickly pear, cactus pear, Barbary fig, cactus fruit, Indian fig, Nopal cactus | Mexico                                                             | Mexico, United States, Italy, Israel, Peru, South Africa, Chile, Australia, Argentina, Colombia, West Indies     | Fruit is ellipsoidal berry, yellow, red, purple or green in color. The fruit length is 10–15 cm long and diameter is 4–5 cm |
| <i>Opuntia streptacantha</i> Lem.     | Nopal cactus, Tray cactus                                                      | Mexico                                                             | Within Mexico                                                                                                    | Round red spiny fruit with bittersweet flavor and 6–7 cm long                                                               |
| <i>Opuntia monacantha</i> Haw.        | Drooping prickly pear, Barbary fig, cochineal prickly pear                     | Northeastern Argentina, Southeastern Brazil, Uruguay, and Paraguay | World widely distributed                                                                                         | Pear shaped, red fruit are thornless and up to 7 cm long                                                                    |
| <i>Opuntia humifusa</i> (Raf.) Raf.   | Devil's tongue, Indian fig, Eastern prickly pear                               | North America                                                      | South Africa, Mauritius, Bulgaria, France, Italy, Spain, Portugal, Brazil, Argentina, New Zealand, Fiji, Uruguay | Juicy red or purple fruit, 3–5 cm long, 6–33 small, flat light-colored seed in each fruit                                   |
| <i>Opuntia phaeacantha</i> Engelm.    | Tulip prickly pear, desert prickly pear, Mohave prickly pear                   | Southwestern United States, northern Mexico                        | Czechia, Slovakia, Spain, United States, Australia                                                               | Edible red or purple fruit with pink seed flesh. Watermelon or pear like flavor and about 5 cm long                         |

segmented by having bright green terminal cladodes that give rise to new growth and flowers. Cladodes can give rise to a few spines (2.5 cm long) or possibly spineless, obovate to oblong 10–40 cm wide and 20–60 cm long. As the plant ages the basal cladodes become woody and the areoles per cladode generally produce only one flower. Yellow or orange cup-shaped flowers are 6–7 cm long and 5–7 cm across (Fig. 1D). The fruit is 5–10 cm long and 4–9 cm across and oblong shaped. Initially the edible fruit is green, turning yellow, orange, red, or purple during ripening (Figs. 1E, 1F) (Janick & Paull, 2008).

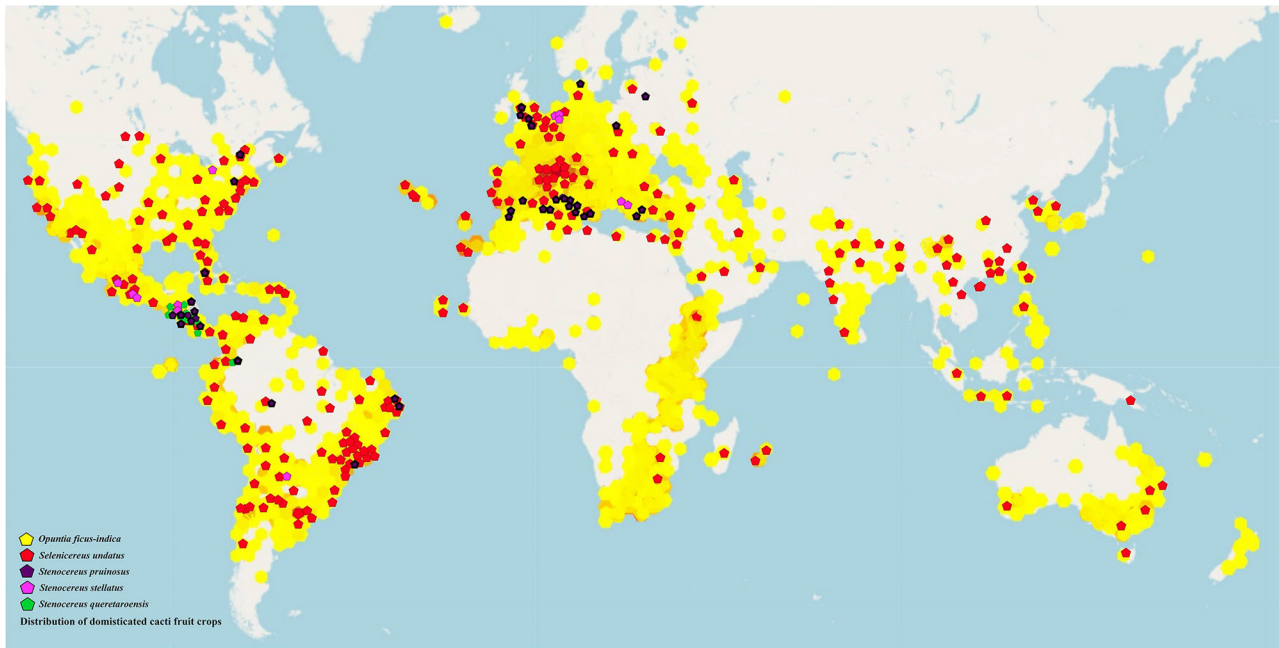
*Opuntia ficus-indica* is globally distributed (Fig. 3) and is famous as a commercial fruit and fodder crop. Australia, Eritrea, Ethiopia, South Africa, and the United States have a high density of *O. ficus-indica*, whereas the species is distributed to a limited extent in the deserts of Pakistan, Somalia, and Yemen (Brutsch & Zimmermann, 1995). Evidence suggests that the species has been cultivated throughout Mesoamerica from pre-Columbian times. Most ethnobotanists believe that *O. ficus-indica* is native to the Central Mexican valley and later introduced around the globe to all continents except Antarctica. The species was first introduced to Europe in the 1500s (Donkin, 1977) and some countries have cultivated the species for more than 100 years. Countries such as Brazil (Dos Santos Sá et al., 2018), Pakistan (Islam et al., 2017), and China (Chen et al., 2021b) have recently started cultivating the species as a food and medicinal crop.

*Opuntia ficus-indica* grows in tropical to subtropical climates, with normal summer rainfall and cold, dry winters (Barbera et al., 1995). Currently the species is successfully farmed in various Mediterranean countries which require irrigation during moisture shortages. The plant can tolerate various climate conditions and can grow in hot summers above 40°C and cold winters below 0°C with average annual rainfall ranging from 250 to 1200 mm (Le Houérou, 2000). The plant does not grow in tropical environments with constant high humidity, in shady areas or during extreme drought periods. The species can grow in natural grasslands, scrub or shrub lands, along railway or roadsides, urban or preurban areas, deserts, and arid regions.

### 6.2.3 Poorly known cultivated species

Most of the well known cacti fruit crops belong to *Selenicereus* and *Opuntia*. Their native regions are Mexico and the United States. Taxonomically, there are 17 *Selenicereus* species, nine of which are found in Mexico: *S. undatus* (Haw.) D.R.Hunt, *S. ocamponis* (Salm-Dyck) D.R.Hunt, *S. stenopterus* (F.A.C.Weber) D.R.Hunt, *S. esculentensis* (Kimmach) D.R.Hunt, *S. guatemalensis* (Eichlam ex Weing.) D.R.Hunt, *S. purpusii* (Weing.) S.Arias & N.Korotkova, *S. calcaratus* (F.A.C.Weber) D.R.Hunt, and *S. triangularis* (L.) D.R.Hunt. Four of the nine are widely distributed: *S. undatus*, *S. purpusii*, *S. triangularis*, and *S. ocamponis* (Fig. 3) (Cáliz, 2004; De Dios, 2005). *Selenicereus costaricensis* (F.A.C.-Weber) S.Arias & N.Korotkova ex Hammel (previously *S. megalanthus*) is found in Nicaragua, while *S. megalanthus* (K.Schum. ex Vaupel) Moran is found in Colombia and Peru. The most cultivated *Selenicereus* species are *S. monacanthus* (Lem.) D.R.Hunt (previously known as *S. polhyrizu*) and *S. undatus*. *Selenicereus* species have become a major cultivated fruit crop in countries of South and Central America including





**Fig. 3.** Distribution map of major cacti food crops. Data for modified distribution map is collected from (Global Biodiversity Information Facility, 2022a, 2022b, 2022c; Pasiecznik, 2022; Patiño-López et al., 2022; Rojas & Praciak, 2022). Yellow color represents worldwide distribution of *Opuntia ficus-indica*, red represents *Selenicereus undatus*, purple represents *Stenocereus prinosus*, pink represents *Stenocereus stellatus*, and green represents *Stenocereus queretaroensis*.

Colombia, Costa Rica, Guatemala, Mexico, Nicaragua, Peru, and Venezuela and have also been introduced to the Bahamas, Bermuda, Cambodia, China, India, Indonesia, Israel, Malaysia, Philippines, Thailand, the United States (California and Florida), and Vietnam (Nerd et al., 2002b). Commonly all *Selenicereus* species are known as pitaya but have different morphological characteristics.

1. *Selenicereus costaricensis* (F.A.C.Weber) S.Arias & N.Korotkova ex Hammel (syn. *Hylocereus costaricensis* (F.A.C.-Weber) Britton & Rose) is a vine cactus, having waxy white stems and margarine (color close to yellow) flowers; having reddish outer perianth at the tips and the small lobes of the stigma. The scarlet fruits are covered with scales and are ovoid in shape. The fruit size varies, is 10–15 cm in diameter and weighs from 250 to 600 g. The flesh of the fruit is red-purple and contains many tiny black seeds, the texture of the flesh is pleasant and has a good taste.
2. *Selenicereus ocamponis* (Salm-Dyck) D.R.Hunt (syn. *Hylocereus ocamponis*) has larger flowers with more or less reddish outer perianth segments; the inner perianth segments are white whereas the middle perianth segments are golden. The scarlet fruits are oblong and covered with large scales, the length of the fruit is 10–15 cm and the weight is 150–400 g. The flesh of the fruit is red and contains several tiny black seeds, the texture of the flesh is similar to ripe kiwi and the flavor is sweet like a blend of pear and kiwi. The flowers are large funnel-shaped white or rarely red.
3. *Selenicereus monacanthus* (Lem.) D.R.Hunt (syn. *Hylocereus monacanthus*) flowers are long (25–30 cm), the outer perianth segments are short and the tips are reddish in color. The lobes of the stigma are yellowish, the

scarlet fruits are oblong and covered with scales. The fruit is 10–15 cm long and weighs 200–400 g. The flesh is red and contains many tiny black seeds. The texture of the flesh is semifirm, dense and tender and the taste is sweet, delicate, and mildly acidic.

*Opuntia* is a large genus of flat-jointed cacti native to the Western Hemisphere with approximately 144 species with varying degrees of domestication (Reyes-Agüero & Valiente-Banuet, 2006). Species of *Opuntia* are cultivated for different purposes and some are important fruit crops. The most widely cultivated is *O. ficus-indica*. Lesser cultivated species are *O. streptacantha* Lem, *O. monacantha* Haw, *O. humifusa* (Raf.) Raf, *O. phaeacantha* Engelm, *O. engelmannii* Salm-Dyck ex Engelm, *O. cochenillifera* (L.) Mill, *O. aurantiaca* Lindley, *O. stricta* (Haw.) Haw, *O. dillenii* (Ker-Gawl.) Haw, *O. streptacantha* Lem, and *O. tomentosa* Salm-Dyck. All *Opuntia* species have different morphological characteristics.

1. *Opuntia stricta* (Haw.) Haw grows from 50 to 100 cm in height. The stem is segmented, large, flattened, and hairless having five spines on the small, raised bumps. The flowers are bright yellow and the fruits are reddish-purple, egg-shaped, and slightly depressed at the tips.
2. *Opuntia dillenii* (Ker-Gawl.) Haw usually grows to 50–100 cm in height, the stem is segmented, elongated, and flattened. There are no hairs on the segments of the stem. The stem bears large spines numbering one to seven on small raised bumps (i.e., areoles) which are grouped. The fruits are reddish-purple and the flowers are bright yellow.

3. *Opuntia streptacantha* Lem. is tall and sometimes tree-like, ranging from 2 to 4 m in height. The segments of the stem are obovate to orbicular flattened. There are no hairs present on the stem and it bears groups of 20–30 small spines that are white in color. Flowers are yellow cup-shaped and the fruits are yellowish or red having many tufts.
4. *Opuntia monacantha* Haw. is tree-like or shrubby and is usually 2–5 m in height. The segmented stem is flattened and elongated. The stem segments are hairless and have two large spines arising from areoles. Flowers are yellow and the fruit is fleshy, with a reddish-purple color and is approximately 60 mm long. The segments of the stem drop toward the ground during fruiting.
5. *Opuntia tomentosa* Salm-Dyck is tree-like at nearly 6 m in height. Stem segments are elongated, flattened, hairy, and lacking spines. The flowers are orange with reddish markings on the underside and the fruit is fleshy, reddish-purple in color, and oval-shaped with shallow depressed tips.

Mexico has been identified as the richest area for cacti species in the world (Valiente-Banuet et al., 1996). Cacti are the principal components of thorn scrub and tropical deciduous forests, which are common landscapes in Mexico. Archeological studies in the Tehuacan Valley (MacNeish, 1967) and Guila Naquitz, Oaxaca (Flannery, 1986) found the oldest evidence of plant domestication in the Western Hemisphere. In ancient times (before the end of the Roman Empire, AD 476) people used cacti mostly due to their easy accessibility, abundance, and high diversity. In the Tehuacan Valley, nearly 74 cacti species were identified and almost all of them are capable of producing edible products (Casas et al., 1997b). Cacti are mainly cultivated for the production of fruits consumed in fresh and dried forms, as well as being used in making candies, ice creams, jams, jellies, and syrups. All species of cacti fruits are consumed except for the subfamily *Perekioidae* (Bravo-Hollis, 1978a). *Opuntia* and some other columnar cacti stems are utilized by indigenous peoples (Callen, 1967). Domestication of plant species has been associated with the cultivation of specific species given that artificial selection is more likely to occur throughout consecutive generations of harvest and breeding of phenotypes of interest (Harlan, 1992). Domestication of plants may also occur under various management of wild plant populations *in situ* including cacti species (Casas et al., 1997a). Spaniards did not consider cacti an important fruit crop to cultivate (Casas et al., 1999) hence its cultivation is not well described like other plant species. Researchers reported the cultivation of *Opuntia* and other cacti species after its introduction to the Mediterranean region and other Eastern Hemisphere countries in Europe, the Mediterranean region, and some Middle Eastern countries in the 16th century (Mizrahi et al., 2010).

## 7 Domestication History of Major Food Crops of Cactaceae

According to ethnobotanists' research in Mesoamerica's southern Mexico and northern Central America, cacti were cultivated by pre-Columbian communities. Approximately 118 species were used and domesticated by people of

Mesoamerica with nearly 40 species being cultivated widely by current indigenous peoples (Casas & Barbera, 2002). Among these, *Selenicereus undatus* is widely valued across Mexico for its edible fruits and decorative value (Bravo-Hollis, 1978b; Ortiz-Hernandez, 1999). This species is thought to be native to the tropical deciduous woods of Mexico, Central America, Northern America, and the West Indies (Bravo-Hollis, 1978a; Nobel, 2002) and is cultivated in China, Colombia, Ecuador, Guatemala, Indonesia, Mexico, Nicaragua, Peru, and Vietnam for its edible fruits. Recently cultivation has increased in Australia, Israel, Japan, New Zealand, the Philippines, the southwestern United States, and Spain (Nerd et al., 2002b; Nobel, 2002).

*Selenicereus undatus* has expanded widely across several tropical areas of the Americas and the Caribbean via birds and propagation during domestication. The species was initially brought by the Spanish to the Philippines in the 16th century and to China in 1645 (Flora of China Editorial Committee, 2014). In 1830 the species was brought to Hawaii in a shipment of plants from Mexico and soon became cultivated across the islands as ornamentals (Morton, 1987). The French introduced *S. undatus* in the 19th century to the Indian subcontinent and China, which later became a major fruit crop for consumption in Southeast Asia and is currently cultivated in tropical and subtropical regions (Ortiz-Hernandez, 1999; Nerd et al., 2002a; Nobel, 2002). The species was introduced to South Africa as an ornamental plant (Walters et al., 2011) and initially brought to Brazil as a garden plant and grown along the roadside (Taylor & Zappi, 2004; Brasil, 2019). *Selenicereus undatus* has also been found in other parts of the world, such as Isla Más a Tierra (Robinson Crusoe Island) off the coast of Chile; Cuba (Oviedo et al., 2012); the island of Île Grand Terre in New Caledonia (MacKee, 1994); and the island of Niue (Space et al., 2004). Naturalized populations of *S. undatus* have also been found in the Canary Islands as well as in Andalusia (Dana et al., 2005a).

*Opuntia ficus-indica* is the most common and commercially significant cactus crop in Mexico (Barrientos, 1965; Nobel, 2002). *Opuntia* is considered one of the most promiscuous interspecific plants, rivaled perhaps only by *Quercus L.* (Benson & Walkington, 1965; Grant & Grant, 1971b, 1979a; Baker & Pinkava, 1987, 1999; Griffith, 2001, 2003; Pinkava, 2002). The plant's vegetative growth shows evidence of occasional clonal domination in specific regions (Grant & Grant, 1971a; Mandujano et al., 1996). Evidence shows that species of *Opuntia* were used by humans as food 9 000 years before present (Kiesling, 1998).

Early European botanists (e.g., Theophrastus or Pliny) named this cactus species "*Ficus indica*" (Donkin, 1977) and was likely to have been recognized in the written history of the Eastern Hemisphere by the beginning of the 16th century (Donkin, 1977; Casas & Barbera, 2002). *Opuntia ficus-indica* is believed to have been introduced by Columbus when he returned to Lisbon in 1493 (Russell & Felker, 1987; Anderson, 2001) and introduced to the Caribbean by the late 1400s. Historical evidence suggests that *O. ficus-indica* was cultivated in Tlaxcala, Mexico in 1519 (Idell, 1957) and that fruits and shoots were used by the people of southeastern Mexico before the 19th century (Coe, 1994). The Nazca of Peru used *O. ficus-indica* from approximately

100 BC to AD 800 (De Rios & Cardenas, 1980), putting these plants in South America early on (Sejuro, 1990). No evidence currently exists recognizing *O. ficus-indica* as native to the Western Hemisphere, however, the geography and evolutionary origins of this species remain unclear (Britton & Rose, 1963; Bravo-Hollis, 1978a; Benson, 1982; Anderson, 2001). The mysterious origins of *O. ficus-indica* are partly due to the widespread cultivation and distribution both within and outside of the Western Hemisphere (Anderson, 2001), but also due to artificial selection and phenotypic variability of the plants (Benson, 1982). The specific characteristics of *O. ficus-indica* may represent 16th century European beliefs that the species is indigenous to the West Indies (Barbera et al., 1992; Casas & Barbera, 2002). Even though the species was brought first to Europe in the 1500s, its domestication and cultivation for more than 100 years is recorded in many other countries including Australia, Eritrea, South Africa, the United States, and also to some extent in Somalia and Yemen (Brutsch & Zimmermann, 1995). Some countries, such as Brazil, China, India, and Pakistan, have recently started the practice of domestication and cultivation.

Other cacti species, such as *Stenocereus pruinosus* (Luna & Del, 1999), *Stenocereus queretaroensis* (Pimienta-Barrios & Nobel, 1994), and *S. stellatus* (Casas et al., 1997a, 1999) have been domesticated in other regions. In southeast-central Mexico, *S. pruinosus* was domesticated, whereas *S. queretaroensis* was domesticated in west-central Mexico, and *S. stellatus* was domesticated in south-central Mexico. These species are grown in tropical deciduous and thorn scrub wild forests, with *in situ* management also being done for the cultivation (Casas et al., 1999). Archeological information collected from caves in Tehuacan shows that *S. stellatus* has been used and managed for its edible fruits for nearly 5 000 years (MacNeish, 1967; Smith, 1967). Wild populations of *S. stellatus* and *S. pruinosus* are managed *in situ* by preserving desirable phenotypes. The cultures were mostly produced in home gardens, where desired phenotypes were reproduced and novel varieties were introduced by volumetric seedlings. Similarly, *S. queretaroensis* populations linked with pre-Columbian communities have been used consistently in southern Jalisco and Colima for a long period (Benz et al., 1997), with the domestication of this species stemming from the 19th century (Pimienta-Barrios & Nobel, 1994). Currently the species is widely cultivated in Guanajuato and Jalisco.

## 8 The Potential Distributions of Cacti Species Used for Food Including Under Climate Change

Cacti fruit crops are distributed on all continents except Antarctica. They are distributed in various ecosystems ranging from sea level to the high Andes. Semiarid subtropical regions on both sides of the equator have a rich diversity of cacti (Bravo-Hollis, 1978a; Gibson & Nobel, 1986; Nobel, 2002). Many cacti flourish in wet tropical forests, as well as colder regions. Physiological adaptations of cacti, such as changing water content and osmotic pressure, enable cacti to tolerate cold stress (Gibson &

Nobel, 1986; Nobel & Bobich, 2002). The spines (modified leaves) on the stem reduce transpiration due to a minimum surface area which can help the plant survive in hot climates (Rebman & Pinkava, 2001). The vertical cladodes of most species avoid high tissue temperature at midday by lowering light interception compared to a horizontal surface (Nobel, 1978, 1980), allowing plants to tolerate high temperatures. Species with horizontal cladodes intercept more light during midday (Sortibrán et al., 2005), which helps individuals survive in shaded conditions. Undergoing CAM photosynthesis (Andrade et al., 2007), having waxy cuticles 5–30  $\mu\text{m}$  thick (Conde, 1975; Pimienta-Barrios et al., 1992; North et al., 1995), and a small number of stomata (20–30) per square millimeter on the stem (Conde, 1975; Pimienta-Barrios et al., 1992) reduce water loss and enable survival in extreme drought conditions. Another adaptation of cacti is in their root system, having extremely fibrous roots allowing maximum water absorption during rainfall. Cacti also have the ability to grow new roots within 24 h after the soil receives moisture (Kausch, 1965), which helps supporting plants grow on shallow and rocky substrates. *Selenicereus* species are lithophytes or hemi-epiphytes and can survive in shady regions and can grow in dry tropical and lowland tropical deciduous forests. They grow naturally in deciduous tropical forests, semideciduous tropical forests, coastal forests, riparian vegetation, thorn scrub, and thorn forests (Arias, 1997; Pérez-García et al., 2001). They are also found in rocky areas, along roadsides and in maritime scrubs (Taylor & Zappi, 2004; Flora of China Editorial Committee, 2014). Individuals may be found in highly varied habitats ranging from 2 to 2 750 m above sea level, with annual average temperatures ranging from 13°C to 29°C and rainfall ranging from 340 to 3 500 mm (Cálix, 2004). *Opuntia* species thrive in temperate to subtropical climates with average summer rainfall and cold, dry winters (Barbera et al., 1995). They show tolerance to changing temperatures and can grow in various climatic conditions with average annual rainfall ranging from 250 to 1 200 mm, with warmer summers above 40°C and colder winters falling below 0°C (Le Houérou, 2000). *Opuntia* species have little to no tolerance to salinity and may tolerate salt saturation in soil up to 5–6 dS/m (Le Houérou, 1996), which impacts growth and development.

## 9 Pollination Biology and its Role in Crop Development

Cactaceae have developed diverse pollination syndromes with their floral characteristics which attract several pollinator species (Proctor et al., 1996). Traits include color, size, and shape of the flower, as well as nectar composition and blooming. Pollination depends on the function of pollinators via a type of pollen and nectar collection and how they pollinate the flower (Fenster et al., 2004). Pollinators pollinate three-quarters of the world's crops (Klein et al., 2007), with approximately 3%–8% of the global food production occurring through animal pollination (Aizen et al., 2009) worth US\$235–577 billion each year (Potts et al., 2016). Animal pollination increases temporal stability of crop yield and shelf life (Klatt et al., 2014) and improves the quality of the crop (Bishop et al., 2022). Not all visitors to

flowers are good pollinators (e.g., nitidulid beetles), with many species having highly specific and few pollinator visits to the flowers for successful pollination (Johnson & Steiner, 2000). In cacti, there are several pollination syndromes mostly represented by four groups of animals, for example, bats (Valiente-Banuet et al., 1996; Tschapka et al., 1999; Fleming & Valiente-Banuet, 2002), bees (Grant & Grant, 1979b; Simpson & Neff, 1987; Schlindwein & Wittmann, 1997; Reyes-Agüero & Valiente-Banuet, 2006; Roig-Alsina & Schlumpberger, 2008), sphingid moths (Silva & Sazima, 1995; Locatelli & Machado, 1999a; Raguso et al., 2003; Schlumpberger & Raguso, 2008; Schlumpberger et al., 2009), and birds (Locatelli & Machado, 1999b; Nassar et al., 2007). Some cacti use multiple pollination vectors in their reproductive strategy, such as *Pachycereus marginatus* (DC.) Britton & Rose (Dar et al., 2006) and northern region populations of *Pachycereus pecten-aboriginum* (Engelmann ex Watson) Britton & Rose which are pollinated by birds and bats.

While vertebrates pollinate cacti around the tropics, a decrease in pollinator diversity has been recorded away from the tropics, in both north and south directions, and at higher elevations (Fleming et al., 2001). Similar observations have been recorded in South America toward the southern limits of the distribution of columnar cacti, where species are pollinated by representatives of different nocturnal and diurnal visitor guilds (Schlumpberger & Badano, 2005; Walter, 2010). Western Hemisphere nectar-feeding bats visit and pollinate night-blooming cacti flowers (Valiente-Banuet et al., 1996; Locatelli et al., 1997; Martins et al., 2016). The cactus flora of the southwestern United States is dominated by bee-pollinated flowers, with 39%–44% of the species possessing large flowers and 50%–56% possessing medium and small flowers (Grant & Grant, 1979a, 1979b). Pollination of cacti by sphingids and hummingbirds is less than pollination by bees and bats. Most of the temperate zone cacti between southern Mexico and northern Panama are pollinated by hummingbirds (Scobell & Scott, 2002).

### 9.1 Abiotic environmental interactions

Environmental abiotic stresses such as high soil salinity, drought, high variability in temperature, ultraviolet radiation, heavy metal accumulation, and high light intensity are the main challenges to global food production and food security (Godfray et al., 2010; Cook et al., 2014; Lesk et al., 2016).

Desert cacti can tolerate some of the highest temperatures and their thick stems exhibit considerable divergence from air temperature during the daytime. For instance, cactus stems can be 10–22 °C above air temperature, resulting in tissue temperatures of up to 65 °C (MacDougal & Working, 1921). Cacti have relatively high heat tolerance and a small surface-to-volume ratio which results in inefficient heat dissipation by convection (Gates & Benedict, 1963). Extensive high temperatures can impact both tissue viability and nocturnal acid accumulation, the latter being an indication of CO<sub>2</sub> uptake by CAM photosynthesis (Smith et al., 1984). High temperatures and heat waves result in a significant reduction in crop yield across the world (Sanghera et al., 2011; Thornton et al., 2014). Higher salinity decreases the growth of roots and shoots of cacti and can

cause damage (Schuch & Kelly, 2008). Salinity can affect the productivity of crops in agriculturally irrigated land by 33% and by 20% of total cultivation worldwide (Shrivastava & Kumar, 2015).

### 9.2 Drought stress

Drought is a critical threat to products in agricultural systems in arid and semiarid regions. Drought frequency will rise by more than 20% by the end of the 21st century, relative to present rates due to higher climate variability (Lobell et al., 2011; Cook et al., 2014). Water scarcity can result in a decrease in crop productivity and crop yield by approximately 40%. There are three major mechanisms that limit crop yield due to soil water deficiency: (i) decrease in radiation-use efficiency, (ii) reduction in harvest index and (iii) reduction in photosynthetically active radiation canopy absorption (Earl & Davis, 2003). Cacti and other succulent plants show resistance to drought stress, by performing CAM photosynthesis to reduce water loss. Cacti needles and spines help the plant survive drought by being able to collect water droplets from moisture in the air or fog (Ju et al., 2012) and transport it via trichomes on the epidermis to be stored in stems filled with mucilage (Nobel, 2003; Kim et al., 2017). The cacti rhizosheath enables the roots to absorb water quickly in large amounts during intermittent rain (Huang & Nobel, 1992) and the cells in the stem cortex are adapted to store water for long periods, preventing evaporation as a provision against long drought stress (Lopez & Nobel, 1991; North & Nobel, 1992; Huang & Nobel, 1993; Nobel & Bobich, 2002).

### 9.3 Ongoing and future development of cacti as fruit crops

Cacti increasingly serve as an agricultural crop, as well as an industrial crop, including fruit, vegetable, and animal feed. Cacti have the greatest potential as a horticultural crop producing unique fruit by many species in various genera. In 1992, the Food and Agricultural Organization of the United Nations (FAO) established the “cactus pear network” to promote the cactus pear as an important fruit crop worldwide. Some *Selenicereus* and *Opuntia* species are now well known for their delicious fruits and their cultivation is promoted. However, most cacti species are not sufficiently described and more work is needed to develop them as fruit crops or for medicinal uses.

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## Supplementary Material

The following supplementary material is available online for this article at <http://onlinelibrary.wiley.com/doi/10.1111/jse.13042/supinfo>:

**Fig. S1.** Phylogenetic tree of *ndhF* gene sequence of species of Cactaceae, Basellaceae, Didieraceae, Aizoaceae, Photocaceae, Nyctaginaceae, Molluginaceae, and Amaranthaceae of Order Caryophyllales. Modified from Wallace & Gibson (2002).